A Purely Analytic Aggregate-Tail Theorem

Abstract

We assemble three independently proved analytic base-frequency inequalities with the curvature propagation from the spherical-shell argument. The result replaces the aggregate interval certificate: for $7/51 \le \rho \le 39/50$ and $K \ge 200$, the continuous reserve is strictly positive and the discrete aggregate implication proves the Dirichlet Pólya inequality. The inclusion $\rho_c = 1/(1+2\pi) > 7/51$ then gives the exact ratio range required in the main proof. No interval enclosure, numerical sign test, or certificate output is used.

1 Definitions and analytic inputs

For $0 \le s \le 1$, put

$$G_1(s) = \frac{\sqrt{1-s^2} - s \arccos s}{\pi} = \frac{1}{\pi} \int_s^1 \arccos t \, dt.$$

For $0 < \rho < 1$, define

$$\eta_\rho = G_1\left(\max\{\rho, \tfrac{1}{2}\}\right), \qquad d_\rho = \frac{2 \arcsin \rho}{\pi}, \qquad b_\rho = \frac{2\pi\rho}{1-\rho}.$$

For $K > 0$, let

$$\mu = \rho K, \qquad \overline{R} = \mu + \frac{1}{2}, \qquad S = \sqrt{\overline{R}^2 + b_\rho K},$$

$$\overline{\mathcal{I}}(\rho, K) = \frac{1}{2}\left[\overline{R}S + b_\rho K \operatorname{arsinh} \frac{\overline{R}}{\sqrt{b_\rho K}} - \overline{R}^2\right], \tag{1}$$

and

$$\begin{aligned}\mathcal{B}(\rho, K) = (\mu - \tfrac{1}{2})(K\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) \\ - (1 + d_\rho)\overline{\mathcal{I}}(\rho, K) - \frac{8(\mu + \frac{1}{2})}{15\sqrt{\mu}}.\end{aligned} \tag{2}$$

Finally, define the curvature floor

$$F(\rho) = 2\rho\eta_\rho + d_\rho \rho^2 - \frac{3(1 + d_\rho)b_\rho}{800}. \tag{3}$$

The three local analytic inputs. The following statements have complete hand proofs in the indicated local notes. They are the only facts imported from those files.

source	imported conclusion on $7/51 \le \rho \le 39/50$
`B-positive.tex`	$\mathcal{B}(\rho, 200) > 60$
`BK-positive.tex`	$\mathcal{B}_K(\rho, 200) > 0$
`F-positive.tex`	$F(\rho) > 1549/84150 > 0$

Each source uses only elementary inequalities and finite exact-rational tables. In particular, none of the three conclusions depends on an interval certificate.

The discrete aggregate input from the main analytic argument. Let

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\},$$

and let $N_D(A_{\rho,1}, K^2)$ count its Dirichlet eigenvalues strictly below K^2. When $\mu > 1/2$, write uniquely

$$\mu - \frac{1}{2} = J + \tau, \qquad J \in \mathbb{N}_0, \qquad 0 \le \tau < 1,$$

and set

$$R = \begin{cases} J, & \tau = 0, \\ J+1, & \tau > 0. \end{cases} \qquad C = b_\rho K.$$

For $C > 0$, define

$$\mathcal{I}(C,R) = \frac{1}{2}\left[R\sqrt{R^2+C} + C \operatorname{arsinh}\frac{R}{\sqrt{C}} - R^2\right]$$

and

$$\mathcal{Q}(\rho,K) = R\lfloor K\eta_\rho \rfloor + d_\rho \frac{J(J+1)}{2} - (1+d_\rho)\mathcal{I}(C,R) - \frac{8R\tau^{5/2}}{15\sqrt{\mu}}. \tag{4}$$

The already-proved discrete aggregate lemma is the analytic implication

$$\mathcal{Q}(\rho,K) \ge 0 \quad \Longrightarrow \quad N_D(A_{\rho,1},K^2) < W(\rho,K) := \frac{2(1-\rho^3)K^3}{9\pi}. \tag{5}$$

Its proof is the concave-head/convex-tail summation and weighted layer-cake argument; it contains no finite computation. We reproduce below the short continuous-to-discrete comparison needed to invoke it.

Lemma 1 (Continuous reserve implies the discrete reserve)**.** *If $\mu > 3/2$, $K\eta_\rho > 1$, and $\mathcal{B}(\rho,K) \ge 0$, then*

$$\mathcal{Q}(\rho,K) > \mathcal{B}(\rho,K) \ge 0.$$

Proof. Put $y = K\eta_\rho > 1$. The strict floor inequality gives

$$R\lfloor y \rfloor > R(y-1) \ge (\mu - \tfrac{1}{2})(y-1).$$

The definition of J, τ, R also gives

$$\frac{d_\rho J(J+1)}{2} > \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}).$$

Indeed, with $x = \mu - 1/2 = J + \tau > 1$, the function $x \mapsto x(x-1)$ is increasing. If $\tau > 0$, then $x(x-1) < J(J+1)$; if $\tau = 0$, the difference is $2J > 0$. The function $x \mapsto \sqrt{x^2+C} - x$ is positive, and

$$\mathcal{I}(C,R) = \int_0^R (\sqrt{x^2+C} - x)\, dx.$$

Since $R < \mu + 1/2 = \overline{R}$,

$$\mathcal{I}(C,R) < \mathcal{I}(C,\overline{R}) = \overline{\mathcal{I}}(\rho,K).$$

Finally $R\tau^{5/2} < \overline{R}$. Substitution of these four inequalities in (4) gives exactly $\mathcal{Q}(\rho,K) > \mathcal{B}(\rho,K)$. □

2 Derivative identities and curvature propagation

For fixed ρ, abbreviate $I(K) = \overline{\mathcal{I}}(\rho, K)$ and

$$P = S - \overline{R}.$$

Differentiating (1) gives

$$I_K = \rho P + \frac{b_\rho}{2} \operatorname{arsinh} \frac{\overline{R}}{\sqrt{b_\rho K}}, \tag{6}$$

$$I_{KK} = \rho^2 \left(\frac{\overline{R}}{S} - 1 \right) + \frac{\rho b_\rho}{2S} + \frac{b_\rho(2\rho K - 1)}{8KS}. \tag{7}$$

Correspondingly,

$$\mathcal{B}_K = \rho(K\eta_\rho - 1) + (\mu - \tfrac{1}{2})\eta_\rho + d_\rho \rho(\mu - 1) - (1 + d_\rho) I_K - \frac{2\rho(2\mu - 1)}{15\mu^{3/2}}, \tag{8}$$

$$E_{KK} = \frac{\rho^2(2\mu - 3)}{15\mu^{5/2}}, \tag{9}$$

$$\mathcal{B}_{KK} = 2\rho\eta_\rho + d_\rho \rho^2 - (1 + d_\rho) I_{KK} + E_{KK}. \tag{10}$$

These are identities, not estimates.

Lemma 2 (Exact validity conditions on the rational superset)**.** *If*

$$\frac{7}{51} \le \rho \le \frac{39}{50}, \qquad K \ge 200,$$

then

$$\mu > \frac{3}{2}, \qquad K\eta_\rho > 1.$$

Proof. First,

$$\mu = \rho K \ge \frac{1400}{51} > \frac{3}{2}.$$

The function G_1 is decreasing because $G_1'(s) = -\arccos(s)/\pi$. Since $\max\{\rho, 1/2\} \le 39/50$,

$$\eta_\rho \ge G_1(39/50).$$

The elementary inequality $\arccos t \ge \sqrt{2(1-t)}$ follows on writing $t = \cos\theta$, since $\sqrt{2(1-t)} = 2\sin(\theta/2) \le \theta$. Combining this with $\pi < 22/7$ gives

$$200\eta_\rho \ge 200 G_1(39/50) > \frac{1400\sqrt{2}}{33} \left(\frac{11}{50} \right)^{3/2}.$$

The square of the final positive quantity is

$$\frac{8624}{225} > 1.$$

Thus $200\eta_\rho > 1$, and $K\eta_\rho \ge 200\eta_\rho > 1$. $\square$

Proposition 3 (Analytic curvature propagation)**.** *For*

$$\frac{7}{51} \le \rho \le \frac{39}{50}, \qquad K \ge 200,$$

one has

$$\mathcal{B}_{KK}(\rho, K) > F(\rho) > \frac{1549}{84150} > 0. \tag{11}$$

Proof. Here

$$S > \overline{R} > \rho K.$$

In (7), the first term is negative. The remaining two terms satisfy

$$\begin{aligned} \frac{\rho b_\rho}{2S} + \frac{b_\rho(2\rho K - 1)}{8KS} &< \frac{\rho b_\rho}{2S} + \frac{2\rho K b_\rho}{8KS} \\ &= \frac{3\rho b_\rho}{4S} < \frac{3b_\rho}{4K} \le \frac{3b_\rho}{800}. \end{aligned}$$

Therefore

$$I_{KK} < \frac{3b_\rho}{800}.$$

Lemma 2 gives $2\mu - 3 > 0$, so $E_{KK} > 0$. Substitution in (10) yields

$$\mathcal{B}_{KK} > 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1 + d_\rho)b_\rho}{800} = F(\rho).$$

The final strict rational bound is precisely the imported conclusion of `F-positive.tex`. □

3 The analytic aggregate-tail theorem

Theorem 4 (Continuous aggregate reserve for all $K \ge 200$)**.** *For every*

$$\frac{7}{51} \le \rho \le \frac{39}{50}, \qquad K \ge 200,$$

one has the quantitative estimate

$$\mathcal{B}(\rho, K) > 60 + \frac{1549}{168300}(K - 200)^2 > 0. \tag{12}$$

Proof. Fix ρ in the stated interval. The imported local lemmas give

$$\mathcal{B}(\rho, 200) > 60, \qquad \mathcal{B}_K(\rho, 200) > 0.$$

Proposition 3 gives

$$\mathcal{B}_{KK}(\rho, s) > \frac{1549}{84150} \qquad (s \ge 200).$$

Twice integrating this strict inequality from 200 to K gives

$$\begin{aligned} \mathcal{B}(\rho, K) &= \mathcal{B}(\rho, 200) + (K - 200)\mathcal{B}_K(\rho, 200) + \int_{200}^{K} (K - s)\mathcal{B}_{KK}(\rho, s)\, ds \\ &> 60 + \frac{1}{2}\frac{1549}{84150}(K - 200)^2, \end{aligned}$$

including $K = 200$ by the strict imported base value. This is (12). □

Theorem 5 (Purely analytic aggregate tail for spherical shells)**.** *Let*

$$\rho_c = \frac{1}{1+2\pi}.$$

For every

$$\rho_c \le \rho \le \frac{39}{50}, \qquad K \ge 200,$$

the Dirichlet counting function of the unit outer-radius spherical shell satisfies

$$N_D(A_{\rho,1}, K^2) < \frac{2(1-\rho^3)K^3}{9\pi}.$$

Proof. The exact Archimedean bound $\pi < 22/7$ gives

$$1 + 2\pi < 1 + \frac{44}{7} = \frac{51}{7}, \qquad \text{hence} \qquad \rho_c > \frac{7}{51}.$$

(Also $\pi > 3$ gives $\rho_c < 1/7 < 39/50$.) Thus the stated ratio interval is contained in the rational superset used in Theorem 4.

On this superset, Lemma 2 verifies $\mu > 3/2$ and $K\eta_\rho > 1$, while Theorem 4 gives $\mathcal{B}(\rho, K) > 0$. Lemma 1 therefore gives $\mathcal{Q}(\rho, K) > 0$. The analytic discrete implication (5) now proves the asserted strict Pólya bound. □

Dependency boundary. The only local-file imports are the three base inequalities listed at the start. The derivative identities, curvature propagation, continuous-to- discrete comparison, rational-domain checks, and two integrations are all displayed in this note. The remaining discrete implication is the previously proved analytic layer-cake lemma from the main manuscript. No aggregate certificate predicate or interval-arithmetic result appears in the dependency chain.